DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
FIRST SESSIONAL
SUBJECT: (CE-513) OPERATING SYSTEMS

Examination	: B.Tech Semester - V	**Seat No.**	:
Date	: 02/08/2022	**Day**	: Tuesday
Time	: 2:30 p.m. to 03:45 p.m.	**Max. Marks**	: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. **[12]**

CO1 (a) Illustrate with example: "Use of multi-programming improves performance of the batch system". [2]

CO2 (b) Which problem may arise in the case of a "five state process model" with a single blocked queue? Explain any solution for the same. [2]

CO2 (c) Name the application where the following approaches of threads and process are implemented: (I) one process one thread (II) one process multiple thread [2]

CO2 (d) Discuss the responsibilities of short term scheduler in uniprocessor scheduling. [2]

CO2 (e) Consider three CPU bound processes which require 8,10,5,6 units for execution and arrive at the time 0,1,2,3 respectively. How many context switches are required if the operating system implements the Round Robin Scheduling algorithm? Do not count context switches at time 0 and at the end. [2] Q = 2

CO2 (f) When does the scheduling algorithm result in the convoy effect? [1]

CO2 (g) Define throughput. [1]

Q.2 Attempt *Any TWO* from the following questions. **[12]**

CO2 (a) Why is the "Not Running" state of "2 states process model" divided further for "5 states process model"? Draw the transitions which occur between Ready, Running and Blocked States for "5 states process model" and support with explanation. [6]

CO2 (b) When does the process switch occur? Differentiate the following mechanisms about interrupting the execution of a process: Interrupt, trap, supervisor call. [6]

CO2 (c) Draw a diagram showing the combined approach of User Level Threads (ULT) and Kernel Level Threads (KLT). Support that the combined approach addresses the issues related with pure ULT and pure KLT. [6]

Q.3 Attempt *Any ONE* from the following questions. **[12]**

CO2 (a) What are the disadvantages of the "Round Robin" algorithm? Illustrate use of "Virtual Round Robin" algorithm with example. [6]

CO2 (b) Consider the following Process with their Burst Time and Arrival Time. The operating system uses a "Round Robin" scheduling algorithm with a time quantum of 2 units to schedule the process. Create a gantt chart to schedule the processes; Find average turnaround time, average waiting time, average response time and CPU idle period in units. [6]

Process_id	Arrival Time	CPU Burst Time (Service Time)
P1	0	4
P2	1	5
P3	2	2
P4	3	1
P5	4	6
P6	5	3

OR

CO2 (a) What is Multilevel Queue Scheduling? Illustrate Multilevel Feedback Queue Scheduling with an appropriate example. [6]

CO2 (b) Consider the following processes with their arrival time and Burst time. Provided the operating system follows the scheduling policy "Shortest Job First" and there is 1 unit of overhead in scheduling the processes, create a gantt chart to schedule the processes; Find the average turnaround time, average waiting time and efficiency of the algorithm. [6]

Process_id	Arrival Time	CPU Burst Time (Service Time)
P1	0	3
P2	1	5
P3	3	2
P4	9	5
P5	12	2